

# 機器視覺檢測

遠心鏡頭及  
線掃描相機

教學講義





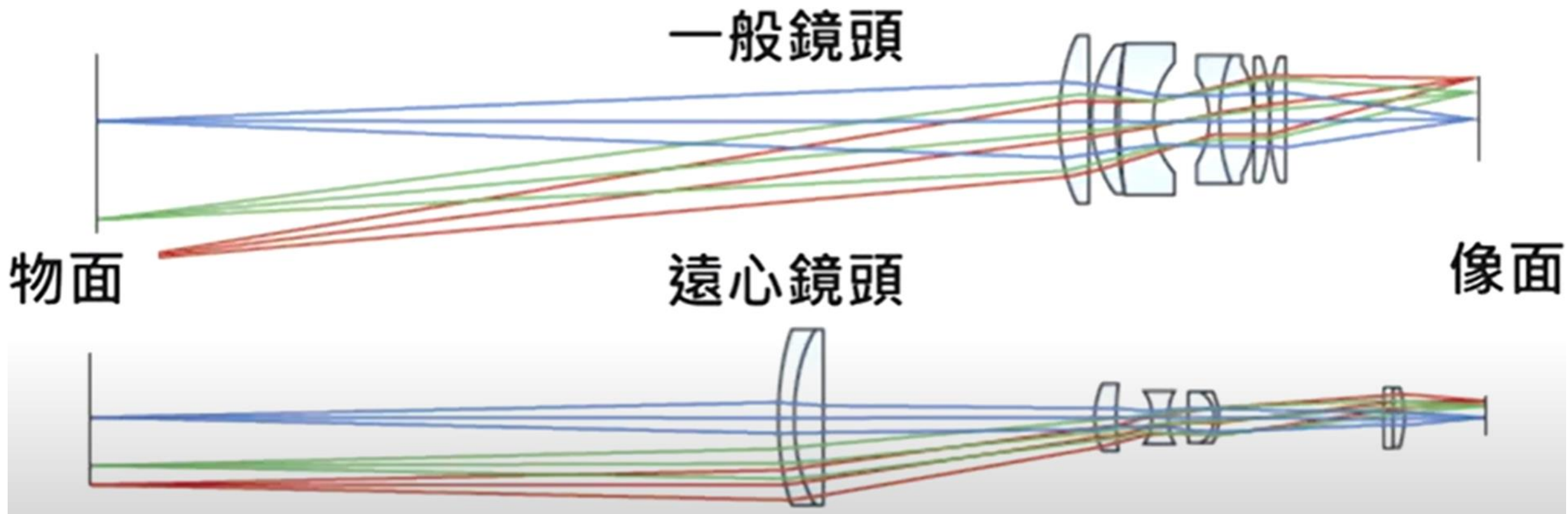
# 單元大綱

■遠心系統

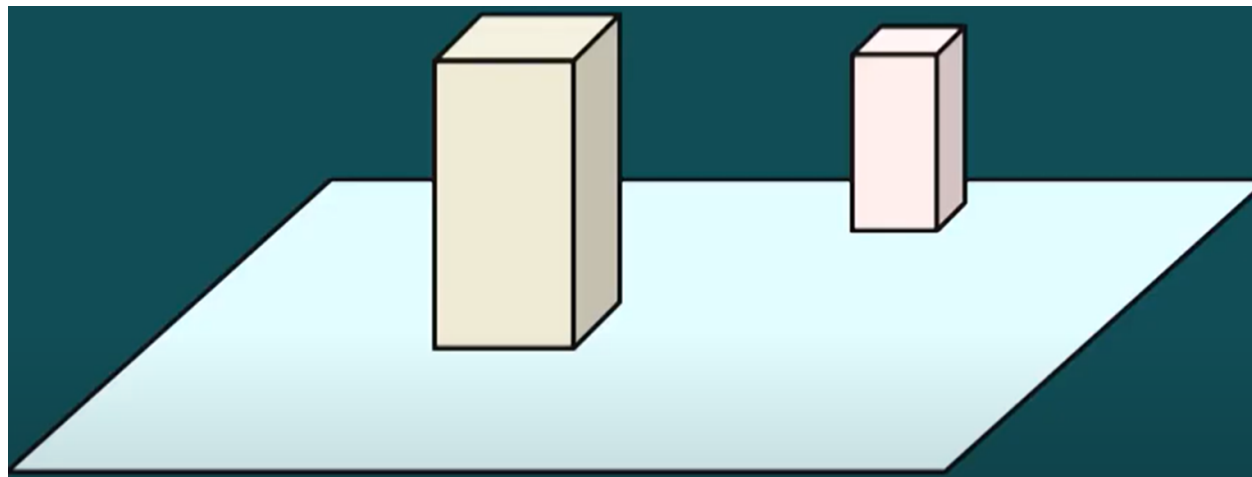
■線掃描系統

# 遠心系統

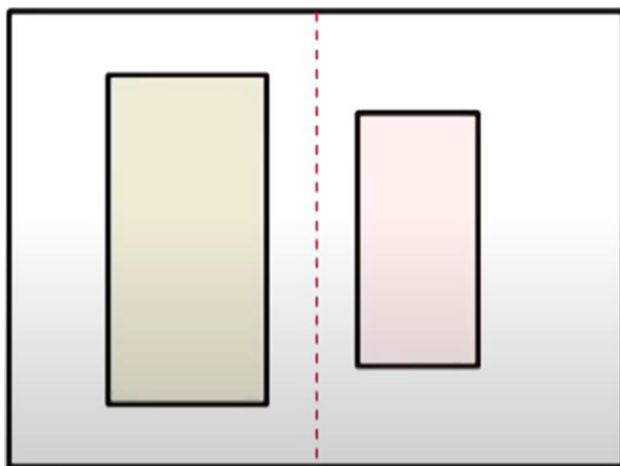
## ■何謂遠心



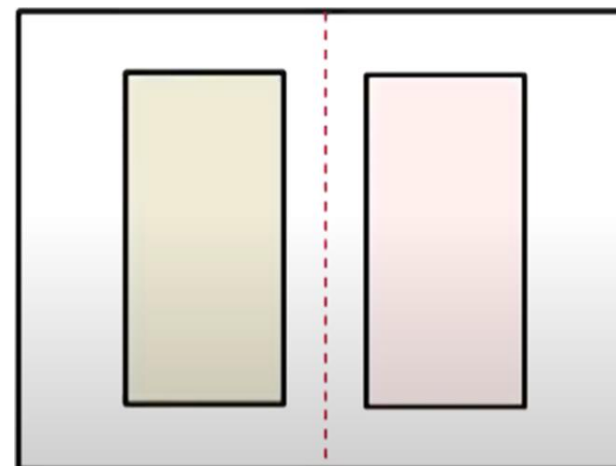
# 遠心系統



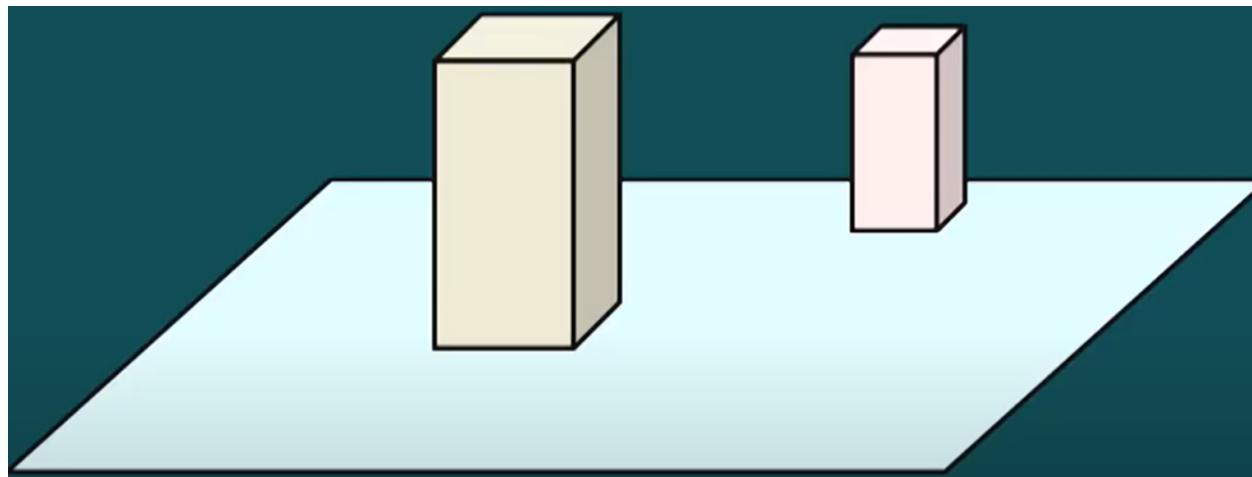
- 一般鏡頭
  - 非等視角



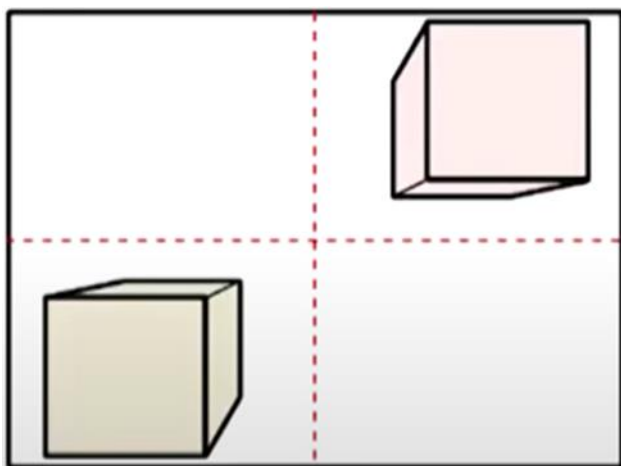
- 遠心鏡頭
  - 等視角



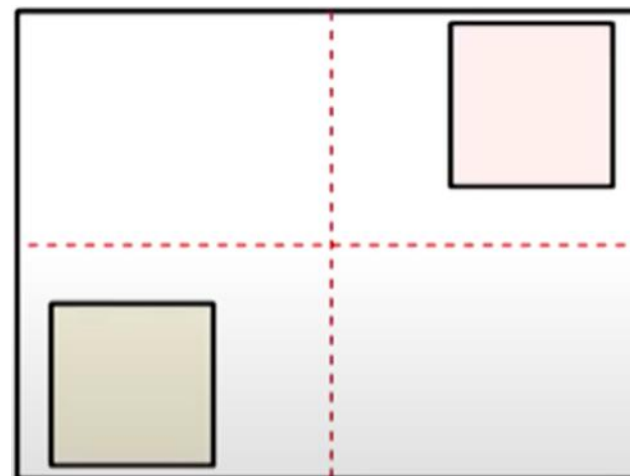
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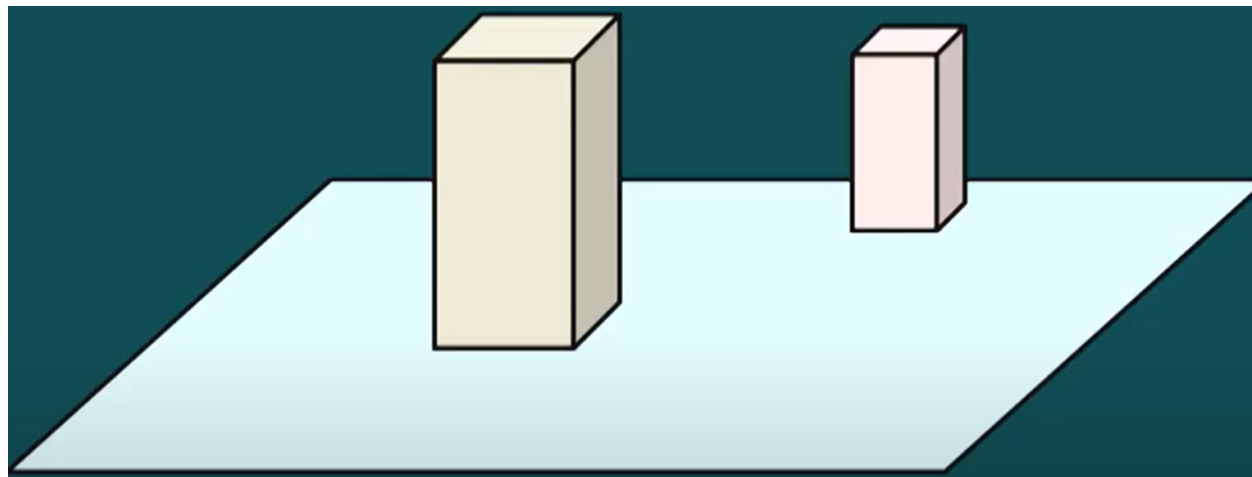
- 一般鏡頭
  - 非等視角



- 遠心鏡頭
  - 等視角

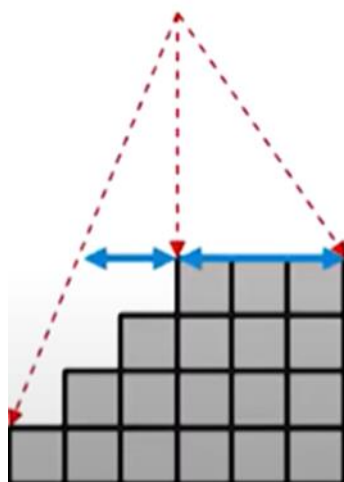


# 遠心系統



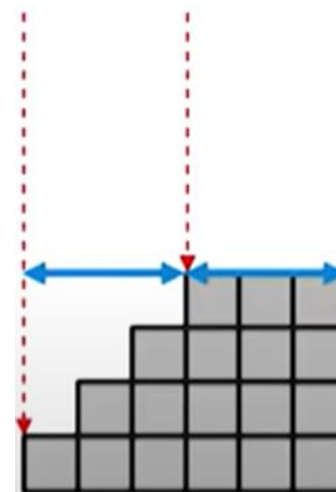
- 一般鏡頭

- 橫向量測會受深度影響



- 遠心鏡頭

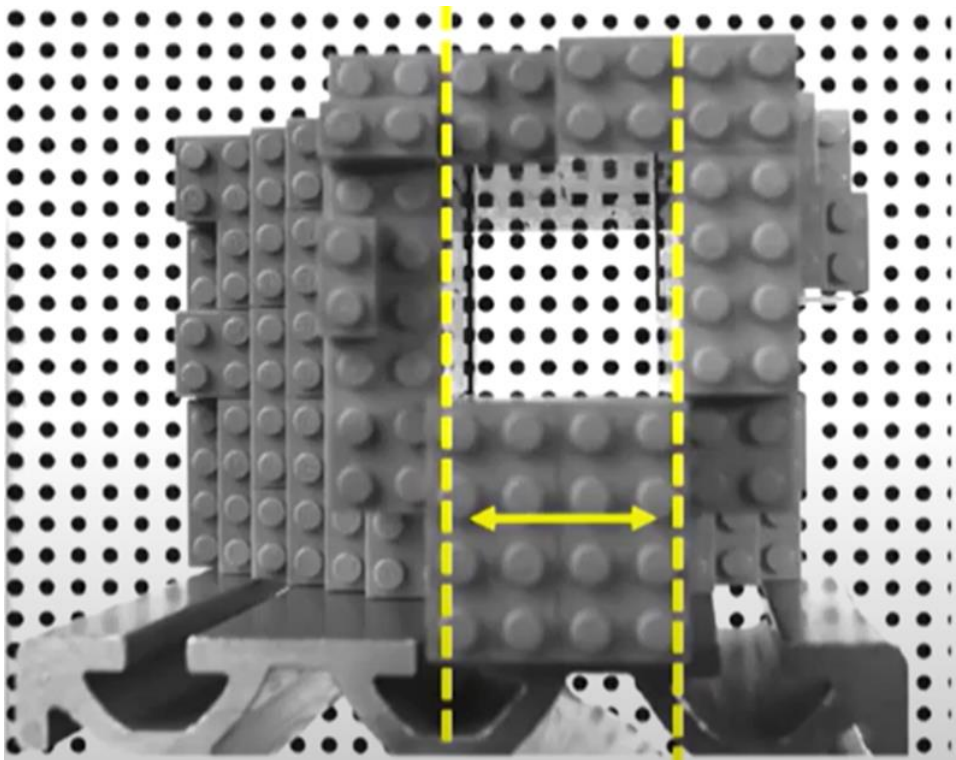
- 橫向量測不會受深度影響



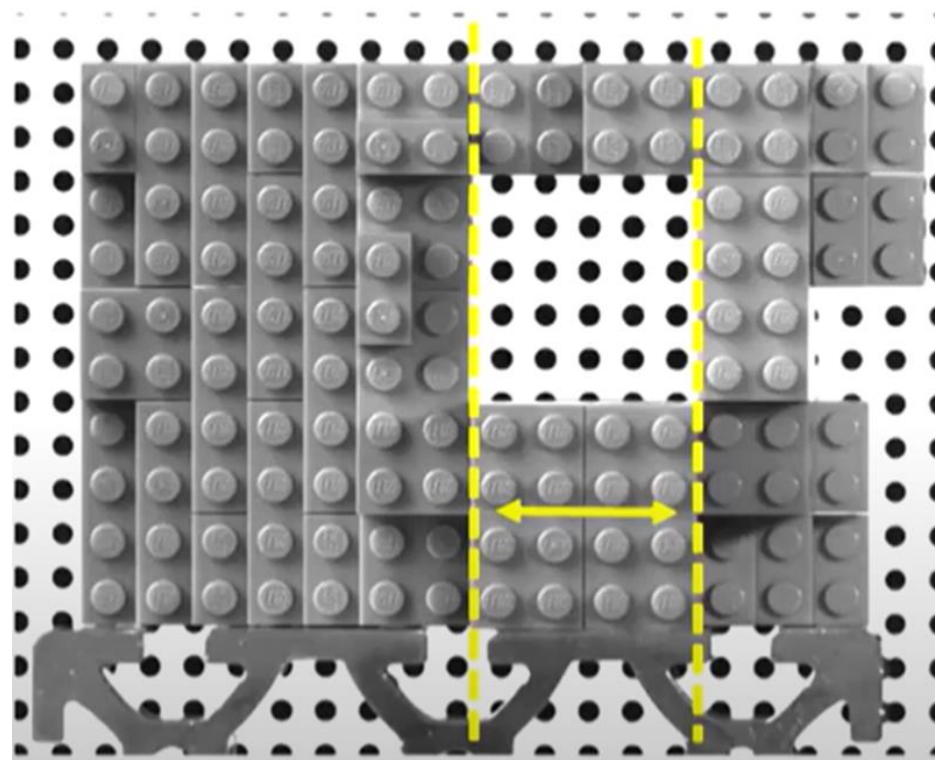


# 遠心系統

## ● 一般鏡頭

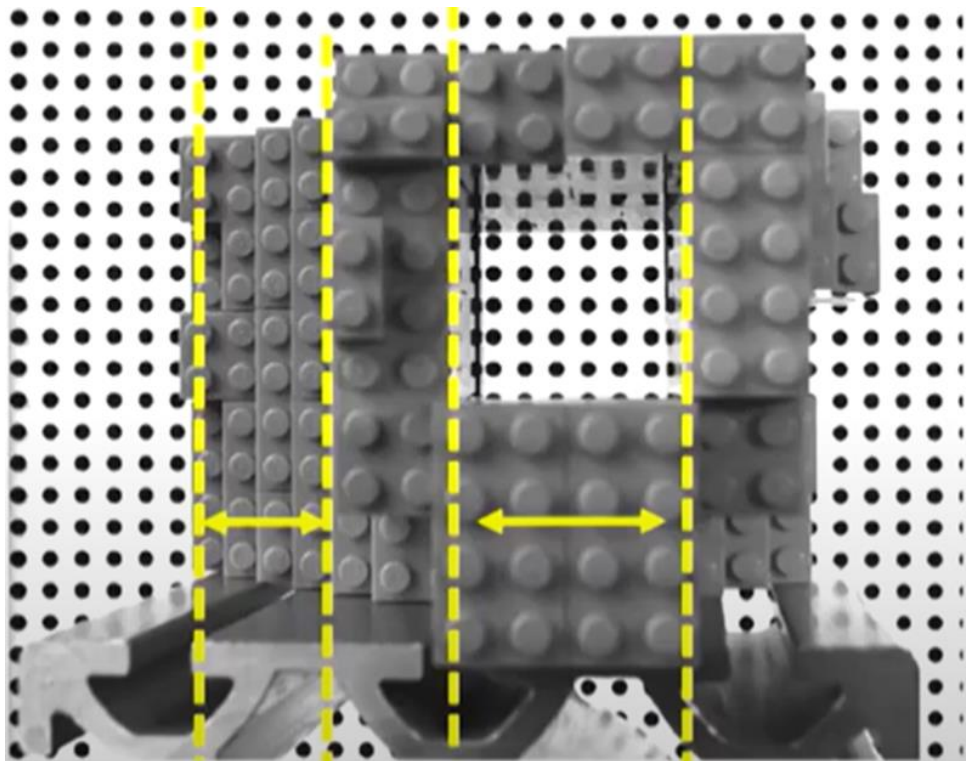


## ● 遠心鏡頭

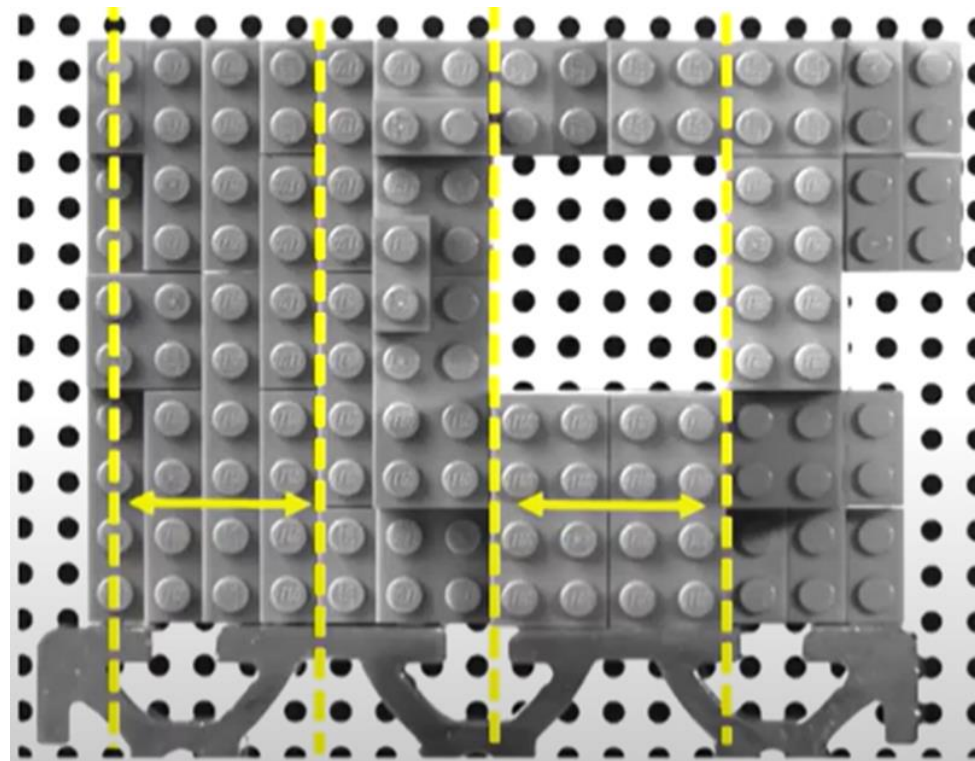


# 遠心系統

## ● 一般鏡頭



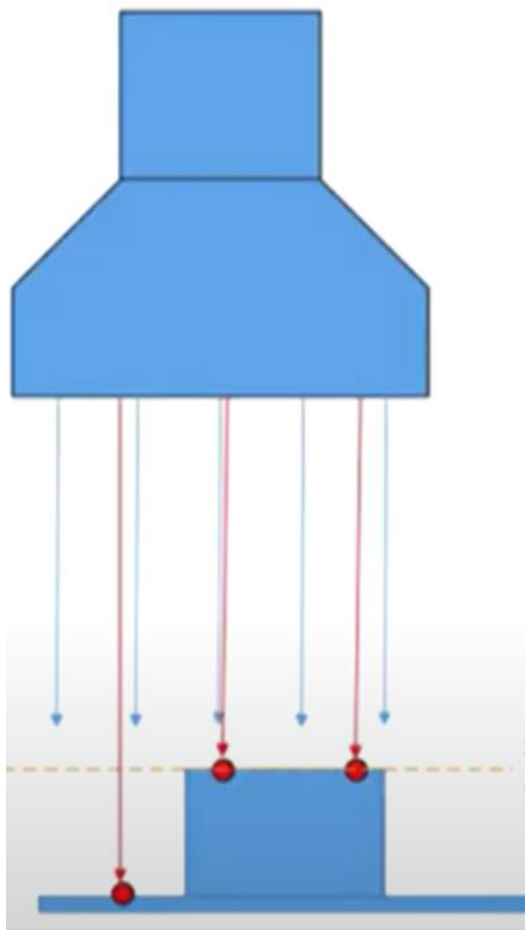
## ● 遠心鏡頭



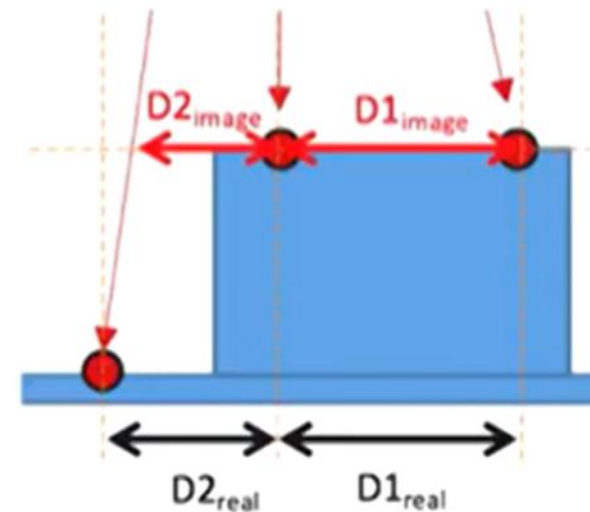
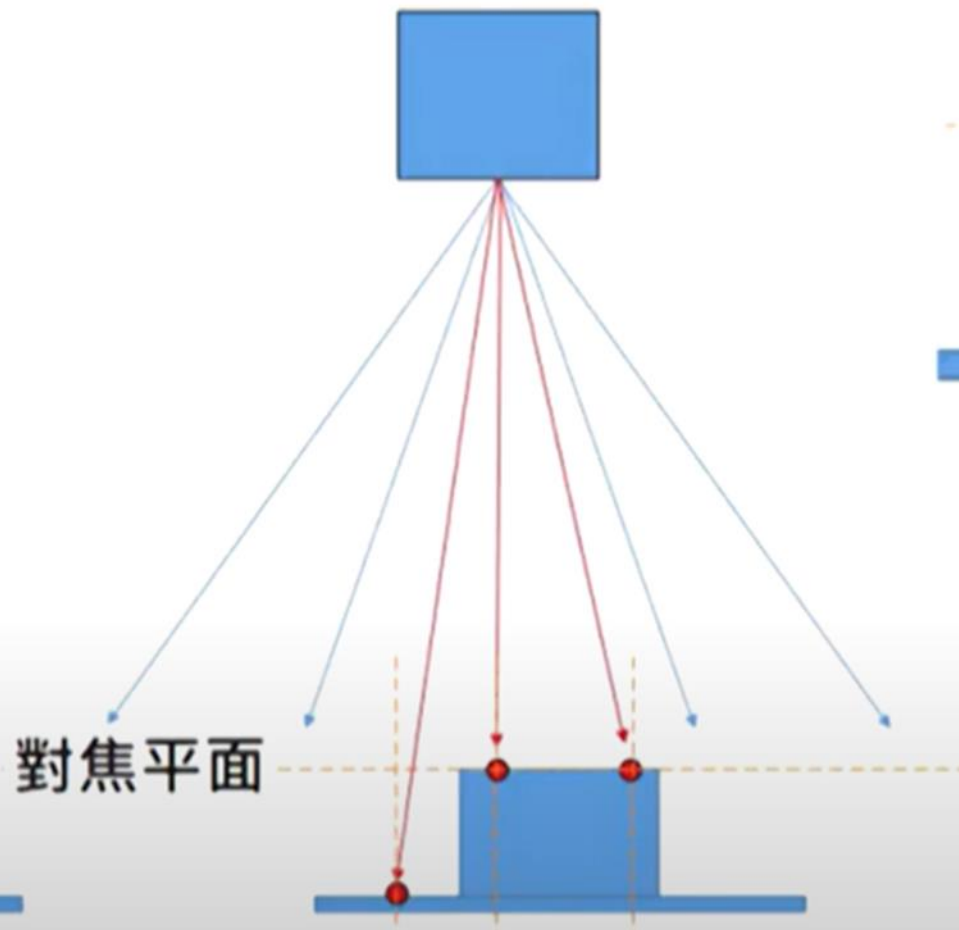


# 遠心系統

- 遠心鏡頭



- CCTV鏡頭



# 遠心系統 - 尺寸量測應用



• CCTV鏡頭

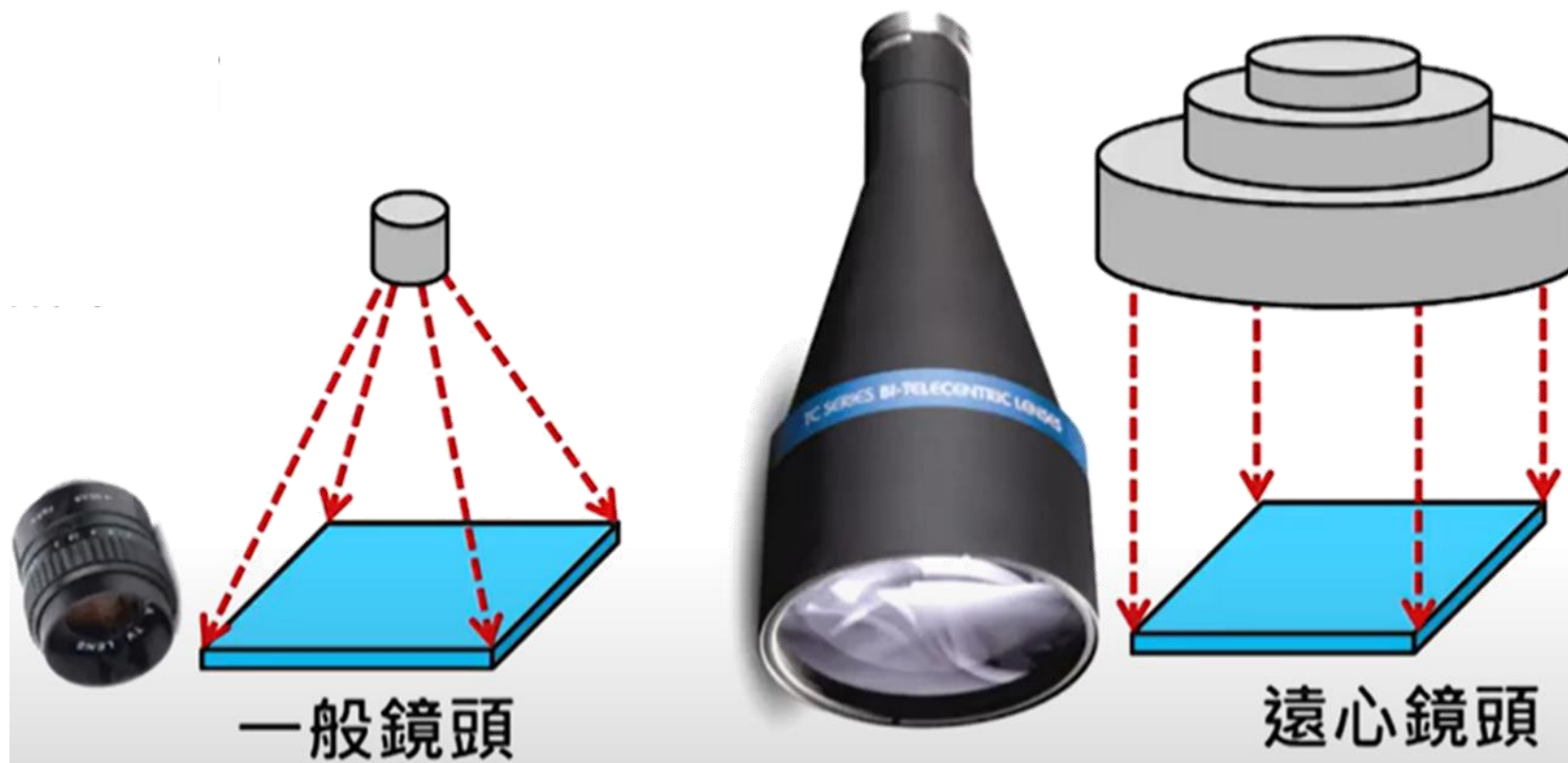


• 遠心鏡頭

# 遠心系統 - 尺寸量測應用

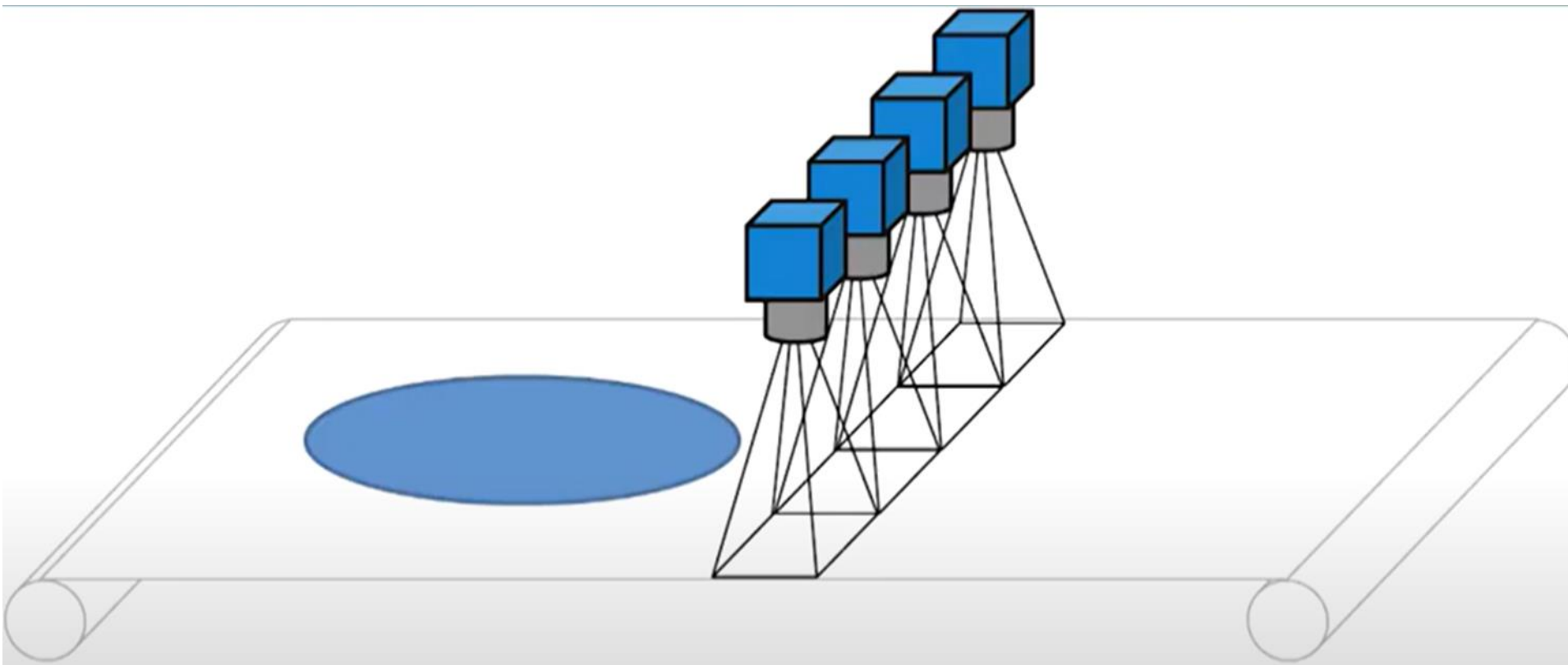
## ■遠心的缺點

- 體積大
- 價格高

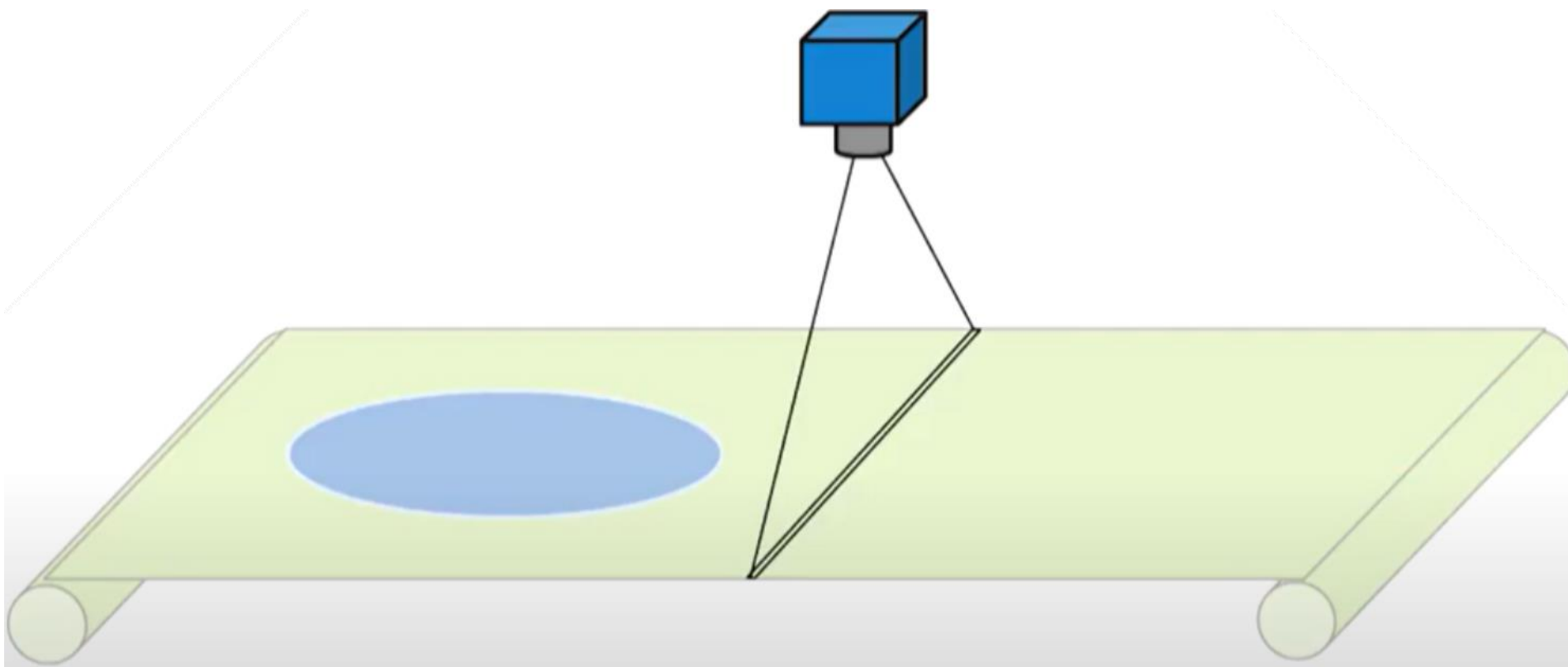


# 線掃描系統 - 面相機陣列

## ■FOV過大時使用



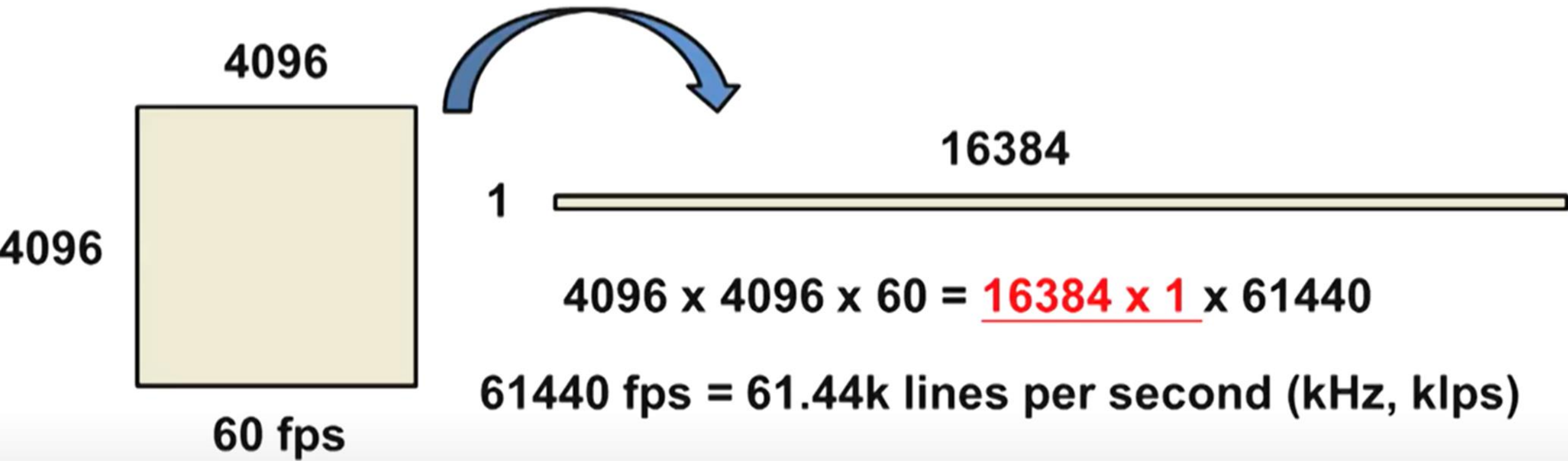
# 線掃描系統 - 線掃描相機



- 60張/秒 4096 x 4096的影像

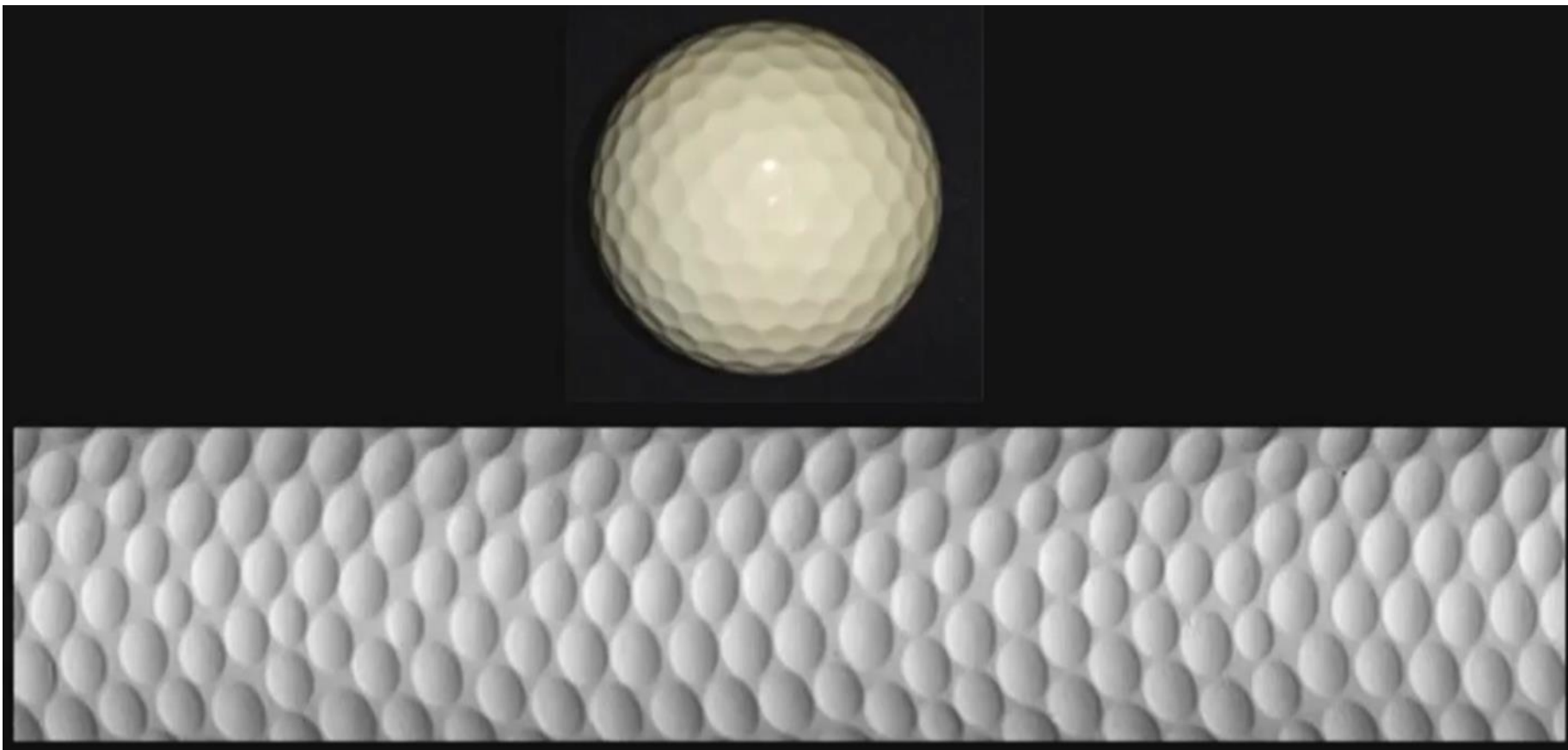


# 線掃描系統 - 線掃描相機



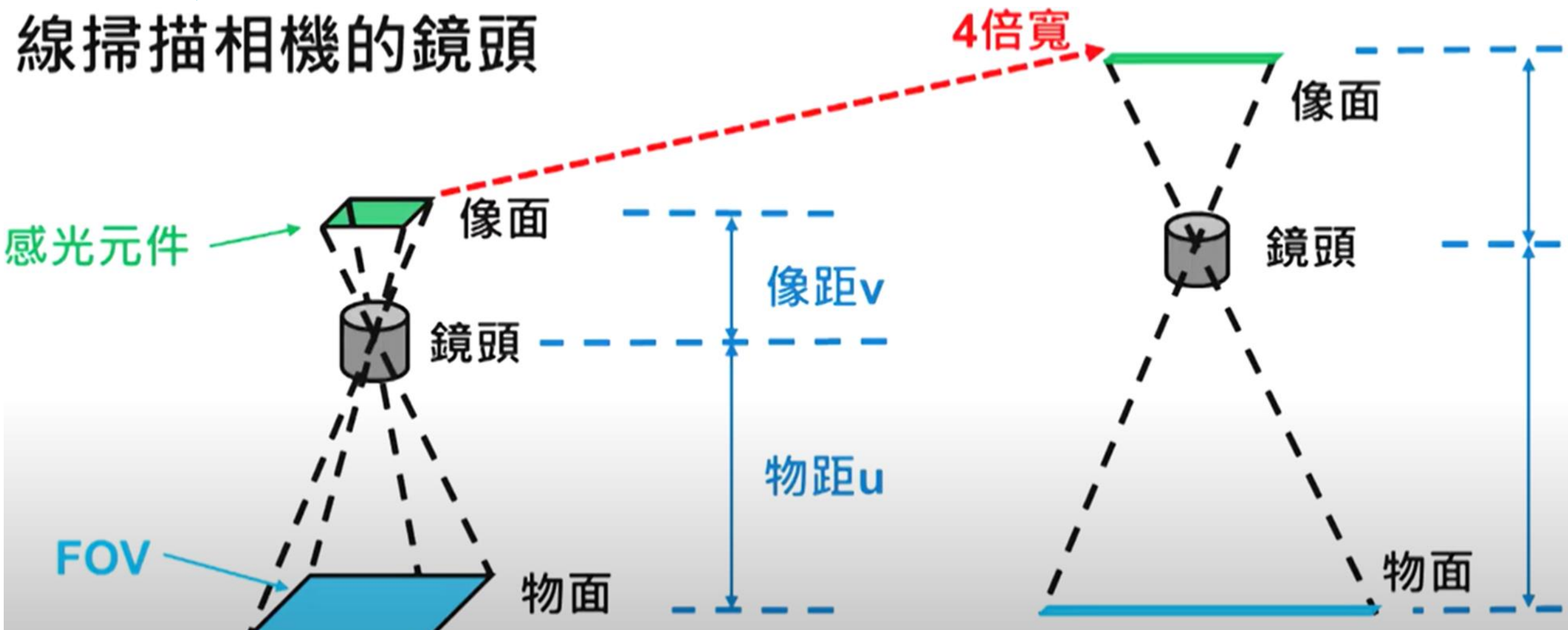
每秒 60 張 4096 x 4096 的影像 → 每秒 61440 張 16384 x 1 的影像

# 線掃描系統 - 線掃描相機



# 線掃描系統

## 線掃描相機的鏡頭



# 線掃描系統

- 線掃描相機的光源
  - 長條狀
  - 高功率、高亮度、高均勻性
  - 60fps
    - 曝光時間=1/60
    - 0.0166s
  - 61440 lines per second
    - 曝光時間=1/61440
    - 0.0000162s→差1000倍

| Series                          | IDBB-LSRF series                                                                    | IDBB-LSRH series                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Product image                   |  |  |
| Illumination method             | Condense light                                                                      | Condense light                                                                      |
| Brightness                      | 1,300,000 lx<br>WD=50mm<br>With condenser lens<br>1,500,000lx                       | 1,000,000 lx<br>WD=50mm<br>With condenser lens<br>1,150,000 lx                      |
| Light-emitting surface (length) | 100mm×n<br>Longest 3,000mm                                                          | 100mm×n<br>Longest 3,000mm                                                          |
| Light control                   | Collectively                                                                        | In 100 mm increments                                                                |
| Air cooling system              | Forced air cooling (fan air cooling)                                                | Natural air cooling                                                                 |



# 註記及感謝

**註記：本講義摘錄自YouTube中DELTA MOOCx  
[科大]二維自動化光學檢測及應用影片，僅  
作為教學輔助使用。**

**感謝台達電子工業股份有限公司製作分享，以及  
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